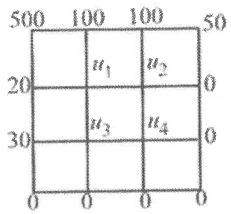
USN | | | | | | | | | | | [24MTAS101]

FIRST SEMESTER M. Tech EXAMINATION, JANUARY - 2025

MATHEMATICS FOR AEROSPACE ENGINEERS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Apply Simpson's (3/8)th rule to evaluate $\int_0^{0.3} \sqrt{1-8x^3}\,dx$ taking seven ordinates	**CO1**	**L3**	**06**
	b.	Solve: $u_{xx}+u_{yy}=0$ in the following square region with the boundary conditions as indicated. 	**CO1**	**L3**	**07**
	c)	Apply Runge-Kutta method of fourth-order to solve $\frac{dy}{dx}=\frac{y^2-x^2}{y^2+x^2}$ with $y(0)=1$ at $x=0.2$ taking h=0.2 as the step length.	**CO1**	**L3**	**07**
2.	**a.**	Solve $x^4\frac{d^3y}{dx^3}+2x^3\frac{d^2y}{dx^2}-x^2\frac{dy}{dx}+xy=\sin(\log x)$.	**CO1**	**L3**	**10**
	b.	A mass of 1 slug stretches a spring 2 ft and comes to rest at equilibrium. The system is attached to a dashpot that imparts a damping force equal to eight times the instantaneous velocity of the mass. Find the equation of motion if an external force equal to f(t)=8sin(4t) is applied to the system beginning at time t=0. What are the transient and the steady-state solution?	**CO1**	**L3**	**10**
3.	**a.**	Define Eigenvalues and Eigenvectors. Diagonalize the matrix $A=\begin{bmatrix}1 & 6 & 1\\ 1 & 2 & 0\\ 0 & 0 & 3\end{bmatrix}$ and hence find A^4.	**CO2**	**L3**	**10**
	b.	Let $\mathbb{R}^+$ be the set of all positive real numbers. Define vector addition as $u+v=uv\ \forall u,v\in\mathbb{R}^+$ and scalar multiplication by $k.u=u^k\ \forall u\in\mathbb{R}^+, k\in\mathbb{R}$.	**CO2**	**L3**	**10**

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		Show that □ $^+$ is vector space over the field of numbers.			
		OR			
4.	**a.**	Show that the vectors v_1= (1,1,2,4), v_2=(2,−1,−5,2), v_3=(1,−1,−4,0), and v_4=(2,1,1,6) are linearly dependent in R^4. Extract a linearly independent subset S from these vectors. Additionally, find the orthogonal subspace S′ to form a basis for R^4 such that S∪S′ spans R^4.	**CO2**	**L3**	**10**
	b.	Define the subspace of vector space and hence check whether the subsets i. $W=\{(x_1,x_2,x_3)/x_1+x_2+x_3=0\}$ of the vector space $V_3(R)$ is a subspace of $V_3(R)$, ii. $W=\{(x_1,x_2,x_3)/x_1^2+x_2^2+x_3^2\le 1\}$ of $V_3(R)$ is a subspace of $V_3(R)$?.	**CO2**	**L3**	**10**
5.	**a.**	Find the inverse Fourier sine transform of $f_s(s)=\dfrac{e^{-as}}{s}$, $a>0$	**CO3**	**L3**	**10**
	b.	Solve one-dimensional heat equation $u_t=c^2u_{xx}$ by the method of separation of variables.	**CO3**	**L3**	**10**
		OR			
6.	**a.**	Find the Fourier series of $f(x)=\begin{cases}1+\frac{4x}{3} & \text{in } \frac{-3}{2}<x<0\\ 1-\frac{4x}{3} & \text{in } 0<x<\frac{3}{2}\end{cases}$ and hence deduce that $\dfrac{1}{1^2}+\dfrac{1}{3^2}+\dfrac{1}{5^2}+\ldots\ldots\ldots=\dfrac{\pi^2}{8}$.	**CO3**	**L3**	**10**
	b.	Solve $\dfrac{\partial^3 z}{\partial x^3}+4\dfrac{\partial z}{\partial x}=0$ given that $z=0$, $\dfrac{\partial z}{\partial x}=0$, $\dfrac{\partial^2 z}{\partial x^2}=4$ when $x=0$.	**CO3**	**L3**	**10**
7.	**a.**	If $\phi+i\psi$ represents the complex potential of an electrostatic field where $\psi=\left(x^2-y^2\right)+\dfrac{x}{x^2+y^2}$, find the complex potential as a function of the complex variable z and hence determine ϕ.	**CO4**	**L3**	**10**
	b.	Use Cauchy's Residue theorem to evaluate the integral $\int_C \dfrac{z^2}{(z-1)(z+2)}dz$ where C is the circle |z| = 2.5.	**CO5**	**L3**	**10**
		OR			
8.	**a.**	Expand $f(z)=\dfrac{1}{(z-1)(z-2)}$ in the region $i)\,\lvert z\rvert<1,\ ii)\ 1<\lvert z\rvert<2$ and $iii)\ \lvert z\rvert>2$	**CO5**	**L3**	**10**
	b.	Evaluate $\int_C \dfrac{\sin z}{\left(z^2-iz+2\right)}dz$ Using Cauchy's integral formula, where C is the rectangle with vertices at (1, 2), (1,3), (-1,3) and (-1, - 2).	**CO5**	**L3**	**10**